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Department of Information Technology

Faculty of Computing

Intake 40

Rapid Application Development (IT2053)

Assignment 02 Part IV Submission

Final Document

Library Management System

Date of Submission: 30th July 2024

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1. Introduction

A Library Management System (LMS) is managing various operations of the library. This system is developed to support the organization and management of library resources. A Library Management System enables the admin to efficiently handle the procurement, cataloging, and inventory of library materials, as well as manage user accounts and facilitate member services by using the concepts of Rapid Application Development (RAD) and implemented using Visual Studio, language C#.

- **Key Features of a Library Management System**

  - **User Authentication and Access Control through the login interface**

- Secure login interface with username and password protection.
- Options for password masking and visibility toggle.
- Error messages for incorrect login attempts to enhance security.
- Single admin access with the ability to add additional admins upon request, ensuring controlled access to system management.

- **Navigation and Interface through the Dashboard interface**

- First interface after the login, helps to have a better navigation about the system.
- Dashboard with buttons for various functions such as adding new books, managing memberships, and other administrative tasks.
- Options to minimize or close the interface for convenience.

- **Book Management through the book management interface**

- update, view, and delete book details.
- Easy navigation back to the dashboard by the back button and logout and exit from the system options.

- **Membership Management through the membership management interface**

- Manage library memberships including adding new members, updating details of a relevant member, viewing a member's information, and deleting members.
- Search functionality to locate specific members using the campus ID.
- Display member details to search and select from the table.
- Easy navigation back to the dashboard by the back button and logout and exit from the system options.

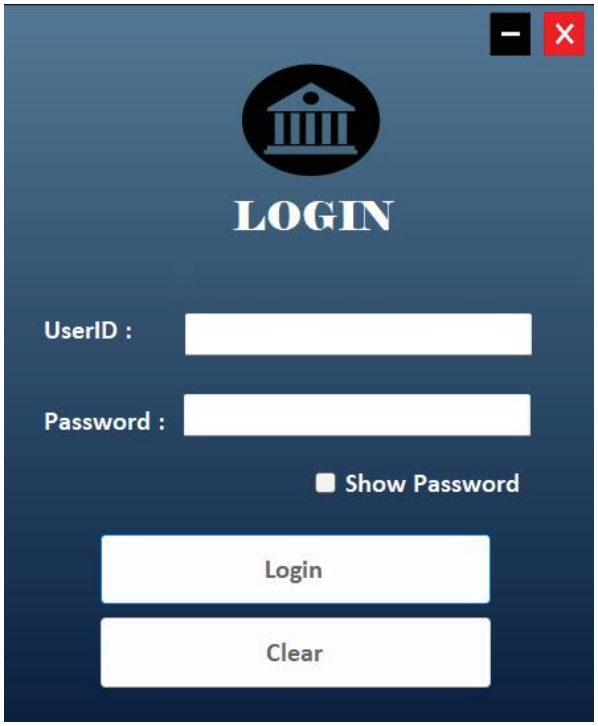
- **Add Books through the New Book Interface**

- Interface to add new book details to the system.

2. System Overview (User Interface, purpose, and the code-related screen)

I. Login Form

- **User Interface**



- **Purpose of this interface**

This is used to secure and prevent the system from unauthorized people making changes to the system. If the entered password or username is incorrect, there will be an error message on a label that will appear, so clear the username and password. If the "show password" tick is clicked that shows the password or else it is displayed in an asterisk (*) mark. There is only one admin in the current system, if any additional person is needed the library should request the administrator to add a new admin this also secures the system.

- **Code**

```
using Microsoft.VisualBasic.ApplicationServices;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using static System.Runtime.InteropServices.JavaScript.JSType;

namespace RAD_Library_Management_System
{
    4 references
    public partial class Login : Form
    {
        1 reference
        public Login()
        {
            InitializeComponent();
            textBox2.PasswordChar = '*';
        }

        SqlConnection con = new("Data Source=DESKTOP-3J22NMI\\SQLEXPRESS;Initial Catalog=LibraryA;Integrated Security=True");

        //Login Button
        1 reference
        private void Btnlogin_Click(object sender, EventArgs e)
        {
            con.Open();
            SqlCommand cmd = new("User_Login", con)
            {
                CommandType = CommandType.StoredProcedure
            };
            cmd.Parameters.Add("@userID", SqlDbType.NVarChar).Value = textBox1.Text;
            cmd.Parameters.Add("@pwsd", SqlDbType.NVarChar).Value = textBox2.Text;
            SqlDataReader dr = cmd.ExecuteReader();
            if (dr.Read())
            {
                Frmhome home = new();
                home.Show();
                this.Hide();
            }
            else
            {
                Invalid.Text = "Invalid login. Please try again.";
                textBox1.Clear();
                textBox2.Clear();
                textBox1.Focus();
            }
            con.Close();
        }
```

```
        //Minimize Button
        1 reference
        private void MinimizeL_Click(object sender, EventArgs e)
        {
            this.WindowState = FormWindowState.Minimized;
        }


        //Close (Exit Application) Button
        1 reference
        private void CLoseL_Click(object sender, EventArgs e)
        {
            Application.Exit();
        }

        //Clear Button
        1 reference
        private void Btnclear_Click(object sender, EventArgs e)
        {
            textBox1.Clear();
            textBox2.Clear();
        }

        //Checkbox to show password in astrics
        1 reference
        private void Show_CheckedChanged(object sender, EventArgs e)
        {
            if (Show.Checked)
            {
                textBox2.PasswordChar = '\0';
            }
            else
            {
                textBox2.PasswordChar = '*';
            }
        }

    }
}
```

```
using Microsoft.VisualBasic.ApplicationServices;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using static System.Runtime.InteropServices.JavaScript.JSType;

namespace RAD_Library_Management_System
{
    public partial class Login : Form
    {
        public Login()
        {
            InitializeComponent();
            textBox2.PasswordChar = '*';
        }

        SqlConnection con = new("Data Source=DESKTOP-3J22NMI\\SQLEXPRESS;Initial
Catalog=LibraryA;Integrated Security=True");

        //Login Button
        private void Btnlogin_Click(object sender, EventArgs e)
        {
            con.Open();
            SqlCommand cmd = new("User_Login", con)
            {
                CommandType = CommandType.StoredProcedure
            };
            cmd.Parameters.Add("@userID", SqlDbType.NVarChar).Value = textBox1.Text;
            cmd.Parameters.Add("@pwsd", SqlDbType.NVarChar).Value = textBox2.Text;
            SqlDataReader dr = cmd.ExecuteReader();
            if (dr.Read())
            {
                Frmhome home = new();
                home.Show();
                this.Hide();
            }
            else
            {
                Invalid.Text = "Invalid login. Please try again.";
```

```
            textBox1.Clear();
            textBox2.Clear();
            textBox1.Focus();
        }
        con.Close();
    }

    //Minimize Button
    private void MinimizeL_Click(object sender, EventArgs e)
    {
        this.WindowState = FormWindowState.Minimized;
    }


    //Close (Exit Application) Button
    private void CLoseL_Click(object sender, EventArgs e)
    {
        Application.Exit();
    }

    //Clear Button
    private void Btnclear_Click(object sender, EventArgs e)
    {
        textBox1.Clear();
        textBox2.Clear();
    }

    //Checkbox to show password in astrics
    private void Show_CheckedChanged(object sender, EventArgs e)
    {
        if (Show.Checked)
        {
            textBox2.PasswordChar = '\0';
        }
        else
        {
            textBox2.PasswordChar = '*';
        }
    }
  }
}
```

II. Dashboard

- **User interface**

- **Purpose of this interface**

This interface opens after the login, it helps to have a better navigation and understanding of the system. By this interface, the admin can click the needed button (add a new book, membership management, etc) and then the system will direct the admin to that relevant page. By clicking the close button, the admin can log out and close the system, and by the minimize button, the admin can minimize the system interface.

- **Codes**

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    5 references
    public partial class Frmhome : Form
    {
        1 reference
        public Frmhome()
        {
            InitializeComponent();
        }

        //Exit (Close) the application
        1 reference
        private void CLose_Click(object sender, EventArgs e)
        {
            Application.Exit();
        }

        //Minimze
        1 reference
        private void Minimize_Click(object sender, EventArgs e)
        {

        }

        //button to go to Add members form
        1 reference
        private void Btnadd_Click(object sender, EventArgs e)
        {
            this.Hide();
            New_book Book = new New_book();
            Book.ShowDialog();
            this.Show();
        }


        //Button to go to Update , delete and view members form
        1 reference
        private void BtnDV_Click(object sender, EventArgs e)
        {
            this.Hide();
            FormBooK Book = new();
            Book.ShowDialog();
            this.Show();
        }

% No issues found
```

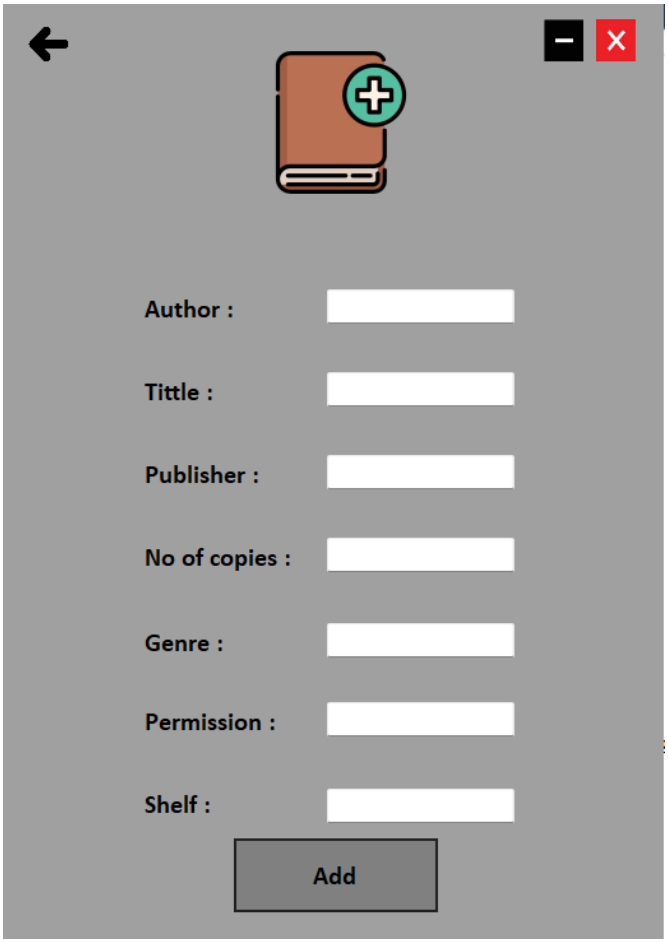
```
        //Button to visit to teh Member management form
        1 reference
        private void Btnmember_Click(object sender, EventArgs e)
        {
            this.Hide();
            FormMemebership member = new FormMemebership();
            member.ShowDialog();
            this.Show();
        }


        //button to visit to  the issue and return books form
        1 reference
        private void BtnIss_Click(object sender, EventArgs e)
        {
            this.Hide();
            FrmIssueReturn issue = new FrmIssueReturn();
            issue.ShowDialog();
            this.Show();
        }

        1 reference
        private void Frmhome_Load(object sender, EventArgs e)
        {

        }
    }
}
```

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    public partial class Frmhome : Form
    {
        public Frmhome()
        {
            InitializeComponent();
        }

        //Exit (Close) the application
        private void CLose_Click(object sender, EventArgs e)
        {
            Application.Exit();
        }

        //Minimze
        private void Minimize_Click(object sender, EventArgs e)
        {
            this.WindowState = FormWindowState.Minimized;
        }

        //button to go to Add members form
        private void Btnadd_Click(object sender, EventArgs e)
        {
            this.Hide();
            New_book Book = new New_book();
            Book.ShowDialog();
            this.Show();
        }

        //Button to go to Update , delete and view members form
        private void BtnDV_Click(object sender, EventArgs e)
        {
            this.Hide();
            FormBooK Book = new();
            Book.ShowDialog();
```

```
            this.Show();
        }

        //Button to visit to teh Member management form
        private void Btnmember_Click(object sender, EventArgs e)
        {
            this.Hide();
            FormMemebership member = new FormMemebership();
            member.ShowDialog();
            this.Show();
        }

        //button to visit to  the issue and return books form
        private void BtnIss_Click(object sender, EventArgs e)
        {
            this.Hide();
            FrmIssueReturn issue = new FrmIssueReturn();
            issue.ShowDialog();
            this.Show();
        }
    }
}
```

III. Add New Book

- **User Interface**



- **Purpose of this interface**

The purpose of this interface is to add books to the system, the admin can add the books and all the text fields should be filled. If the book adding is a success it will display the “Book added successfully.” message in the area below the book image.

By clicking the arrow mark, can go back to the dashboard, by clicking the close sign admin can log out from the system, by the minimize mark the admin can minimize the interface.

- **Code**

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Security.Policy;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    5 references
    public partial class New_book : Form
    {
        1 reference
        public New_book()
        {
            InitializeComponent();
        }

        SqlConnection con = new SqlConnection("Data Source=DESKTOP-3J22NMI\\SQLEXPRESS;Initial Catalog=LibraryA;Integrated Security=True");

        //minimize button
        1 reference
        private void CLoseN_Click(object sender, EventArgs e)
        {
            this.Close();
        }



        //back button
        1 reference
        private void BackN_Click(object sender, EventArgs e)
        {
            this.Close();
        }


        //minimize button
        1 reference
        private void MinimizeN_Click(object sender, EventArgs e)
        {
            this.WindowState = FormWindowState.Minimized;
        }

```

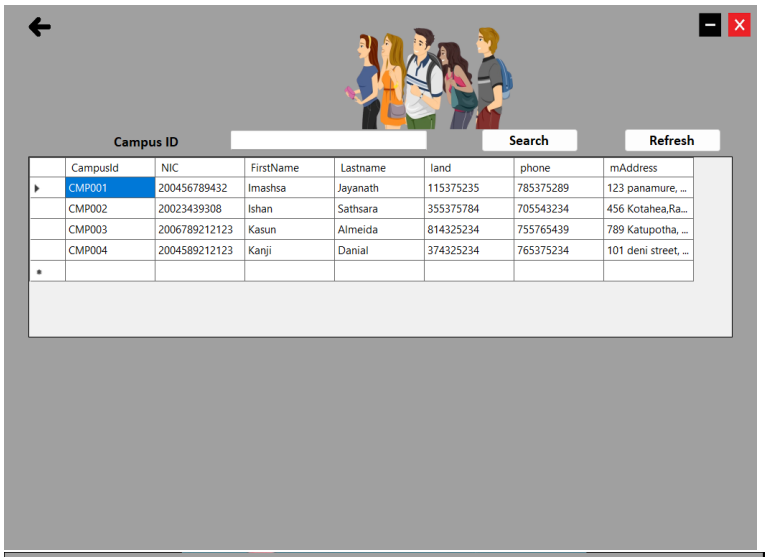
```

        //addnew book button
        1 reference
        private void Btnadd_Click(object sender, EventArgs e)
        {
            con.Open();
            SqlCommand cmd = new SqlCommand("CreateBook", con);
            cmd.CommandType = CommandType.StoredProcedure;
            cmd.Parameters.AddWithValue("@Author", TxtauthA.Text);
            cmd.Parameters.AddWithValue("@Title", TxtTiittleA.Text);
            cmd.Parameters.AddWithValue("@Publisher", TxtpubA.Text);
            cmd.Parameters.AddWithValue("@Genre", TxtGenreA.Text);
            cmd.Parameters.AddWithValue("@Permission", TxtpwerA.Text);
            cmd.Parameters.AddWithValue("@Shelf", TxtshelfA.Text);
            int numberOfCopies;
            if (!string.IsNullOrWhiteSpace(TxtnoA.Text) && int.TryParse(TxtnoA.Text, out numberOfCopies))
            {
                cmd.Parameters.AddWithValue("@NumberOfCopies", numberOfCopies);
                int rowsAffected = cmd.ExecuteNonQuery();
                if (rowsAffected > 0)
                {
                    Lblmsg.Text = "Failed to create book record";
                }
                else
                {
                    Lblmsg.Text = "Book record created successfully.";
                    TxtauthA.Clear();
                    TxtGenreA.Clear();
                    TxtnoA.Clear();
                    TxtpubA.Clear();
                    TxtpwerA.Clear();
                    TxtshelfA.Clear();
                    TxtTiittleA.Clear();
                }
            }
            con.Close();
        }
    }
}
```

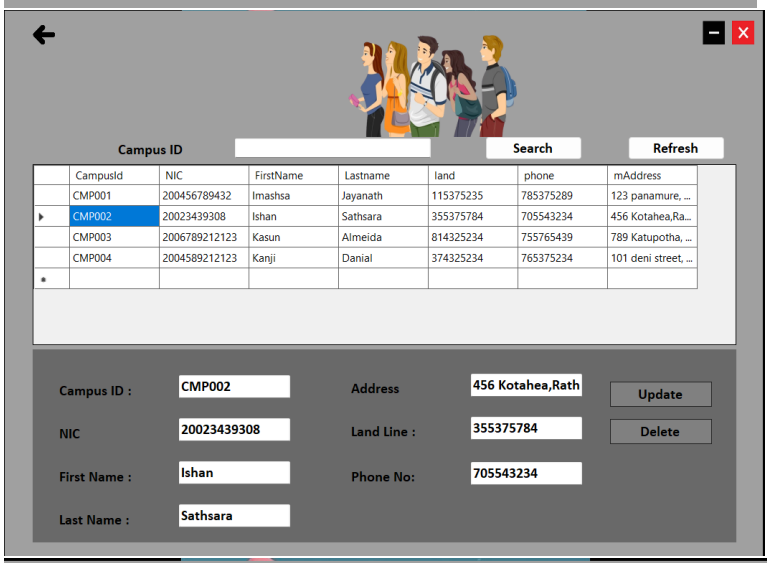
```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Security.Policy;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    public partial class New_book : Form
    {
        public New_book()
        {
            InitializeComponent();
        }

        SqlConnection con = new SqlConnection("Data Source=DESKTOP-
3J22NMI\\SQLEXPRESS;Initial Catalog=LibraryA;Integrated Security=True");

        //minimize button
        private void CLoseN_Click(object sender, EventArgs e)
        {
            this.Close();
        }

        //back button
        private void BackN_Click(object sender, EventArgs e)
        {
            this.Close();
        }

        //minimize button
        private void MinimizeN_Click(object sender, EventArgs e)
        {
            this.WindowState = FormWindowState.Minimized;
        }

        //addnew book button
        private void Btnadd_Click(object sender, EventArgs e)
        {
            con.Open();
```

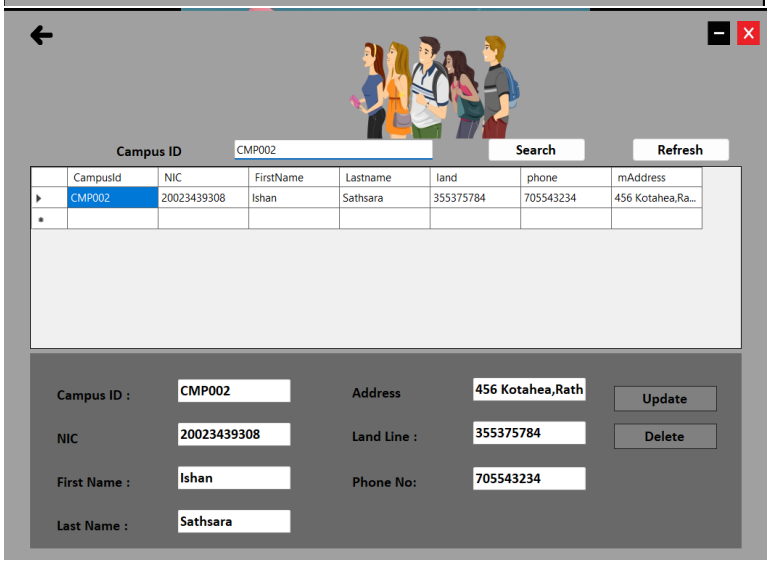
```
            SqlCommand cmd = new SqlCommand("CreateBook", con);
            cmd.CommandType = CommandType.StoredProcedure;
            cmd.Parameters.AddWithValue("@Author", TxtauthA.Text);
            cmd.Parameters.AddWithValue("@Title", TxtTiittleA.Text);
            cmd.Parameters.AddWithValue("@Publisher", TxtpubA.Text);
            cmd.Parameters.AddWithValue("@Genre", TxtGenreA.Text);
            cmd.Parameters.AddWithValue("@Permission", TxtpwerA.Text);
            cmd.Parameters.AddWithValue("@Shelf", TxtshelfA.Text);
            int numberOfCopies;
            if (!string.IsNullOrWhiteSpace(TxtnoA.Text) && int.TryParse(TxtnoA.Text, out
numberOfCopies))
            {
                cmd.Parameters.AddWithValue("@NumberOfCopies", numberOfCopies);
                int rowsAffected = cmd.ExecuteNonQuery();
                if (rowsAffected > 0)
                {
                    Lblmsg.Text = "Failed to create book record";
                }
                else
                {
                    Lblmsg.Text = "Book record created successfully.";
                    TxtauthA.Clear();
                    TxtGenreA.Clear();
                    TxtnoA.Clear();
                    TxtpubA.Clear();
                    TxtpwerA.Clear();
                    TxtshelfA.Clear();
                    TxtTiittleA.Clear();
                }
            }
            con.Close();
        }
    }
}
```

IV. Membership Management

- User Interface







- **Purpose of the screen**

This interface helps with membership management in the library (add new members, update member details, view specific member details, and delete members). When visiting the interface, it appears as the provided first screenshot. We have entered the relevant student's campus ID and click search to search for the specific students' details, so as in the third screenshot after searching the table filter it and display and the details will appear in the panel as shown. Also, we can view a member's detail by clicking the campus ID in the table, the second screenshot shows it.

There also by clicking the arrow mark, can go back to the dashboard, by clicking the close sign admin can log out from the system, by the minimize mark admin can minimize the interface.

- **Code**

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    5 references
    public partial class FormMemebership : Form
    {
        1 reference
        public FormMemebership()
        {
            InitializeComponent();
        }

        SqlConnection con = new SqlConnection("Data Source=DESKTOP-3J22NMI\\SQLEXPRESS;Initial Catalog=LibraryA;Integrated Security=True");

        1 reference
        private void CLoseN_Click(object sender, EventArgs e)
        {
            Application.Exit();
        }

        1 reference
        private void MinimizeN_Click(object sender, EventArgs e)
        {
            this.WindowState = FormWindowState.Minimized;
        }

        1 reference
        private void BackN_Click(object sender, EventArgs e)
        {
            this.Close();
        }
        1 reference
        private void BtnSearch_Click(object sender, EventArgs e)
        {
            string CampusID = TxtSearchID.Text;
            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "SELECT * FROM LMember where CampusId = '" + CampusID + "'";
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
            da.Fill(ds);

            GridMember.DataSource = ds.Tables[0];
            PanelEdit.Visible = true;
        }
```

```
Int64 Landline;
Int64 Phoneno;
1 reference
private void GridMember_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
    try
    {
        if (e.RowIndex >= 0 && GridMember.Rows[e.RowIndex].Cells["CampusId"].Value != null)
        {
            string campusId = GridMember.Rows[e.RowIndex].Cells["CampusId"].Value.ToString();

            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "SELECT * FROM LMember WHERE CampusId = @CampusId";
            cmd.Parameters.AddWithValue("@CampusId", campusId);

            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
            da.Fill(ds);

            if (ds.Tables[0].Rows.Count > 0)
            {
                DataRow row = ds.Tables[0].Rows[0];

                TxtcampusId.Text = row["CampusId"].ToString();
                TxtNIC.Text = row["NIC"].ToString();
                Txtfirst.Text = row["FirstName"].ToString();
                TxtLast.Text = row["Lastname"].ToString();
                TxtLand.Text = row["land"].ToString();
                TxtPhone.Text = row["phone"].ToString();
                Txtaddr.Text = row["mAddress"].ToString();

                PanelEdit.Visible = true;
            }
        }
    }
```

```
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred: " + ex.Message);
    }
}

1 reference
private void BtnRef_Click(object sender, EventArgs e)
{
    TxtSearchID.Clear();
    PanelEdit.Visible = false;
}


1 reference
private void FormMemebership_Load(object sender, EventArgs e)
{

    SqlCommand cmd = new SqlCommand();
    cmd.Connection = con;
    cmd.CommandText = "SELECT * FROM LMember ";
    SqlDataAdapter da = new SqlDataAdapter(cmd);
    DataSet ds = new DataSet();
    da.Fill(ds);

    GridMember.DataSource = ds.Tables[0];
    PanelEdit.Visible = false;
}
```

```
{

    try
    {
        if (MessageBox.Show("Data is going to be updated.Confirm?", "sucess", MessageBoxButtons.OKCancel, MessageBoxIcon.Question) == DialogResult.OK)
        {
            con.Open();

            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "UPDATE LMember SET CampusId = @CampusId, FirstName = @FirstName, Lastname = @Lastname, NIC = @NIC, land = @land, mAddress = @mAddress, phone = @phone WHERE CampusId = @CampusId";
            cmd.Parameters.AddWithValue("@CampusId", TxtcampusId.Text);
            cmd.Parameters.AddWithValue("@FirstName", Txtfirst.Text);
            cmd.Parameters.AddWithValue("@Lastname", TxtLast.Text);
            cmd.Parameters.AddWithValue("@NIC", TxtNIC.Text);
            cmd.Parameters.AddWithValue("@land", TxtLand.Text);
            cmd.Parameters.AddWithValue("@mAddress", Txtaddr.Text);
            cmd.Parameters.AddWithValue("@phone", TxtPhone.Text);

            cmd.ExecuteNonQuery();
            con.Close();
            SqlCommand refreshCmd = new SqlCommand("SELECT * FROM LMember", con);
            SqlDataAdapter da = new SqlDataAdapter(refreshCmd);
            DataSet ds = new DataSet();
            da.Fill(ds);
            if (ds.Tables.Count > 0)
            {
                GridMember.DataSource = ds.Tables[0];
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show(ex.Message);
    }
}
```

```
private void TxtSearchID_TextChanged(object sender, EventArgs e)
{
    if (TxtSearchID.Text != "")
    {
        SqlCommand cmd = new SqlCommand();
        cmd.Connection = con;
        cmd.CommandText = "SELECT * FROM LMember where CampusId Like '" + TxtcampusId.Text + "%'";
        SqlDataAdapter da = new SqlDataAdapter(cmd);
        DataSet ds = new DataSet();
        da.Fill(ds);
        GridMember.DataSource = ds.Tables[0];
    }
}
}
}
```

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    public partial class FormMemebership : Form
    {
        public FormMemebership()
        {
            InitializeComponent();
        }

        SqlConnection con = new SqlConnection("Data Source=DESKTOP-
3J22NMI\\SQLEXPRESS;Initial Catalog=LibraryA;Integrated Security=True");

        private void CLoseN_Click(object sender, EventArgs e)
        {
            Application.Exit();
        }

        private void MinimizeN_Click(object sender, EventArgs e)
        {
            this.WindowState = FormWindowState.Minimized;
        }

        private void BackN_Click(object sender, EventArgs e)
        {
            this.Close();
        }
        private void BtnSearch_Click(object sender, EventArgs e)
        {
            string CampusID = TxtSearchID.Text;
            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "SELECT * FROM LMember where CampusId = '" +
CampusID + "'";
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
```

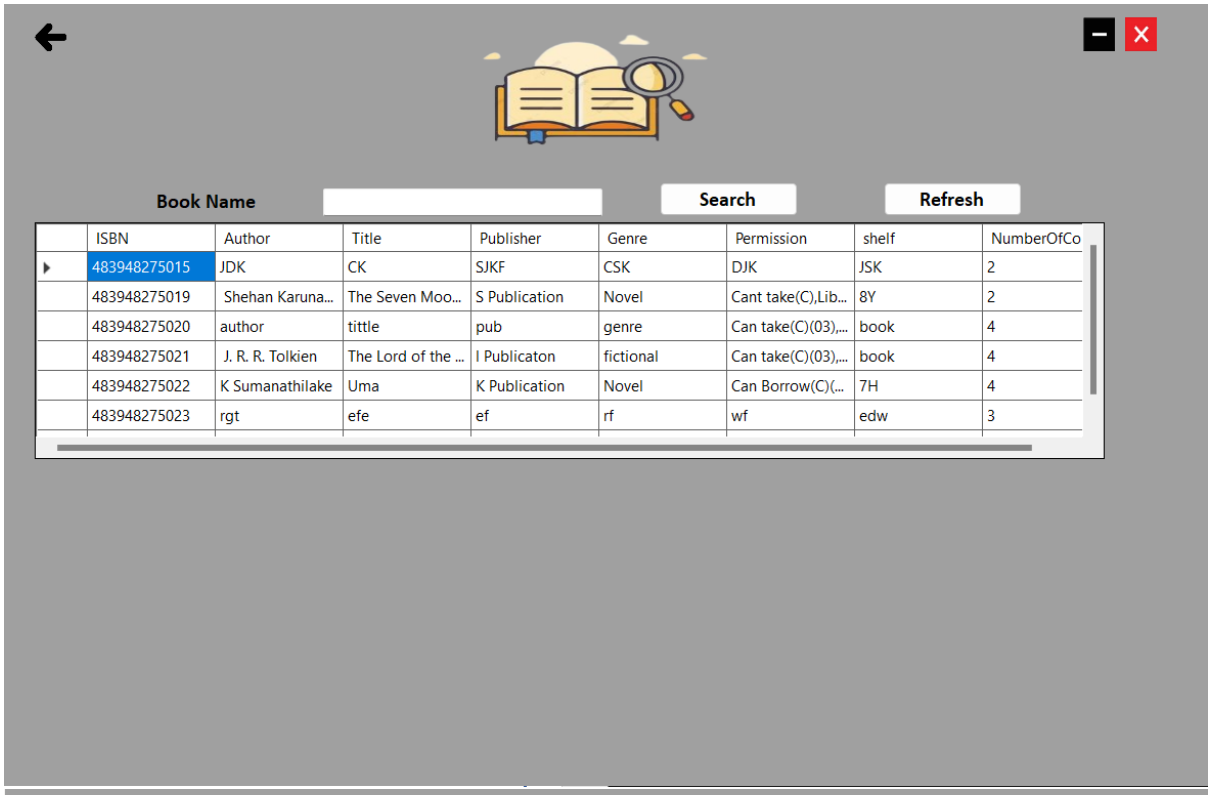
```
            da.Fill(ds);
            GridMember.DataSource = ds.Tables[0];
            PanelEdit.Visible = true;
        }

        Int64 Landline;
        Int64 Phoneno;
        private void GridMember_CellContentClick(object sender,
DataGridViewCellEventArgs e)
        {
            try
            {
                if (e.RowIndex >= 0 &&
GridMember.Rows[e.RowIndex].Cells["CampusId"].Value != null)
                {
                    string campusId =
GridMember.Rows[e.RowIndex].Cells["CampusId"].Value.ToString();
                    SqlCommand cmd = new SqlCommand();
                    cmd.Connection = con;
                    cmd.CommandText = "SELECT * FROM LMember WHERE CampusId =
@CampusId";
                    cmd.Parameters.AddWithValue("@CampusId", campusId);
                    SqlDataAdapter da = new SqlDataAdapter(cmd);
                    DataSet ds = new DataSet();
                    da.Fill(ds);
                    if (ds.Tables[0].Rows.Count > 0)
                    {
                        DataRow row = ds.Tables[0].Rows[0];
                        TxtcampusId.Text = row["CampusId"].ToString();
                        TxtNIC.Text = row["NIC"].ToString();
                        Txtfirst.Text = row["FirstName"].ToString();
                        TxtLast.Text = row["Lastname"].ToString();
                        TxtLand.Text = row["land"].ToString();
                        TxtPhone.Text = row["phone"].ToString();
                        Txtaddr.Text = row["mAddress"].ToString();
                        PanelEdit.Visible = true;
                    }
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show("An error occurred: " + ex.Message);
            }
        }
```

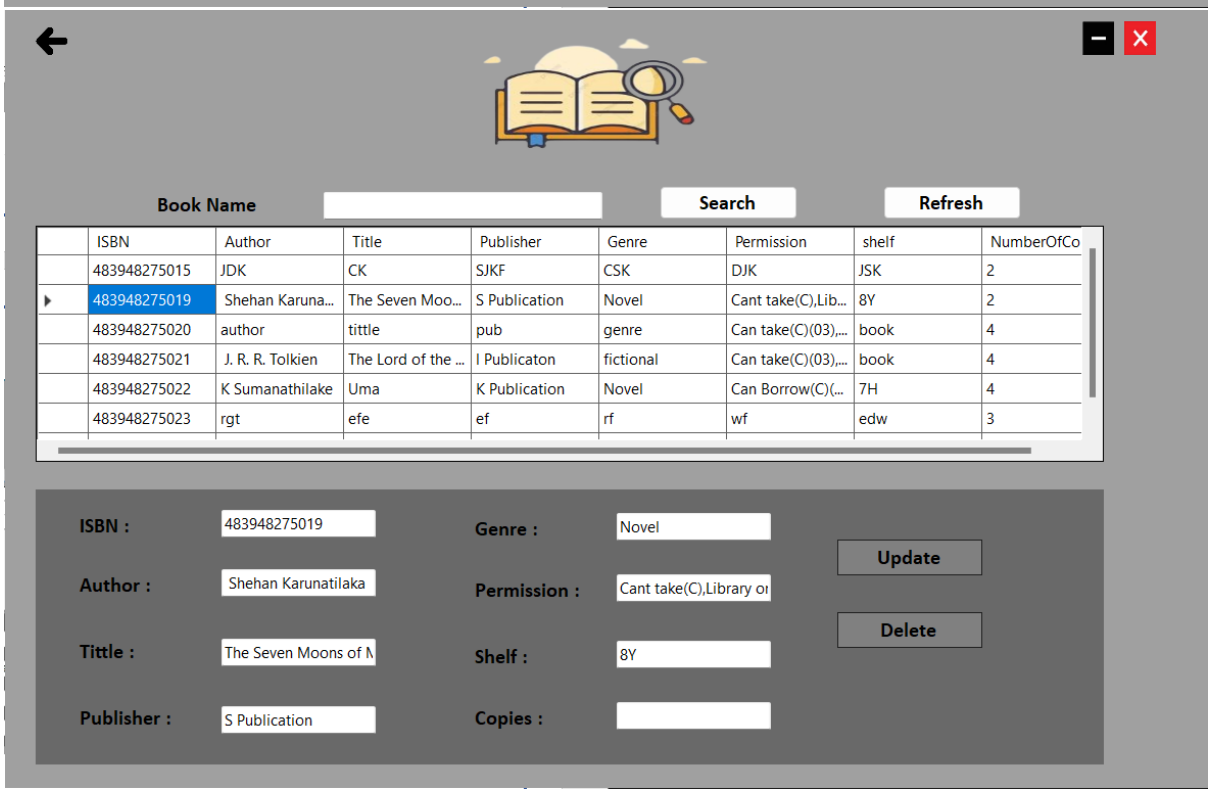
```
        private void BtnRef_Click(object sender, EventArgs e)
        {
            TxtSearchID.Clear();
            PanelEdit.Visible = false;
        }

        private void FormMemebership_Load(object sender, EventArgs e)
        {
            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "SELECT * FROM LMember ";
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
            da.Fill(ds);
            GridMember.DataSource = ds.Tables[0];
            PanelEdit.Visible = false;
        }

        private void Btnupdt_Click(object sender, EventArgs e)
        {
            try
            {
                if (MessageBox.Show("Data is going to be updated.Confirm?", "sucess",
MessageBoxButtons.OKCancel, MessageBoxIcon.Question) == DialogResult.OK)
                {
                    con.Open();
                    SqlCommand cmd = new SqlCommand();
                    cmd.Connection = con;
                    cmd.CommandText = "UPDATE LMember SET CampusId = @CampusId,
FirstName = @FirstName, Lastname = @Lastname, NIC = @NIC, land = @land, mAddress
= @mAddress, phone = @phone WHERE CampusId = @CampusId";
                    cmd.Parameters.AddWithValue("@CampusId", TxtcampusId.Text);
                    cmd.Parameters.AddWithValue("@FirstName", Txtfirst.Text);
                    cmd.Parameters.AddWithValue("@Lastname", TxtLast.Text);
                    cmd.Parameters.AddWithValue("@NIC", TxtNIC.Text);
                    cmd.Parameters.AddWithValue("@land", TxtLand.Text);
                    cmd.Parameters.AddWithValue("@mAddress", Txtaddr.Text);
                    cmd.Parameters.AddWithValue("@phone", TxtPhone.Text);
                    cmd.ExecuteNonQuery();
                    con.Close();
                    SqlCommand refreshCmd = new SqlCommand("SELECT * FROM LMember",
con);
                    SqlDataAdapter da = new SqlDataAdapter(refreshCmd);
                    DataSet ds = new DataSet();
                    da.Fill(ds);
                    if (ds.Tables.Count > 0)
```

```
                {
                    GridMember.DataSource = ds.Tables[0];
                }
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show(ex.Message);
        }
    }

    private void TxtSearchID_TextChanged(object sender, EventArgs e)
    {
        if (TxtSearchID.Text != "")
        {
            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "SELECT * FROM LMember where CampusId Like '" +
TxtcampusId.Text + "%'";
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
            da.Fill(ds);
            GridMember.DataSource = ds.Tables[0];
        }
    }
  }
}
```

V. Book Management (Delete, View, Update)

- Interface

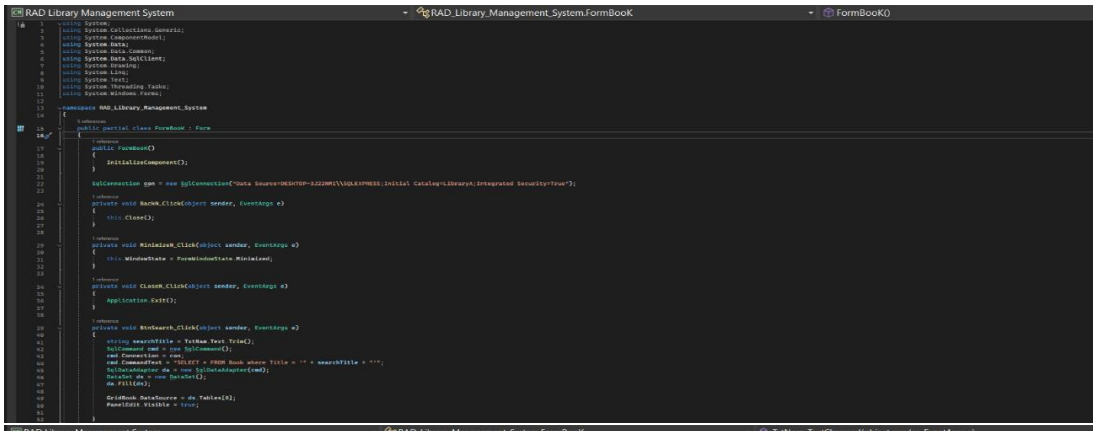


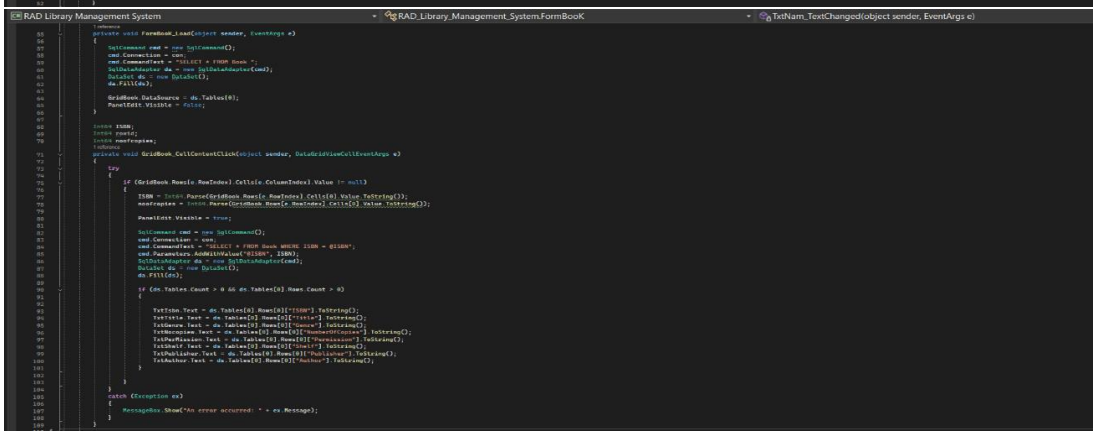


- **Purpose of the screen**

This screen is used to manage book details (update, delete, or view a relevant Book detail). The books are displayed on the table and we can enter the book name in the Book name text box which is on top of the interface. And only after entering the book name and clicking the search button the grey color panel in the second screenshot. By clicking the refresh button admin can clear the relevant searched detail. By using the panel, we can edit(update) or remove the relevant book detail.

- **Code**





```
RAD Library Management System    RAD_Library_Management_System.FormBooK    Btnupdt_Click(object sender, EventArgs e)
111         private void TxtNam_TextChanged(object sender, EventArgs e)
112         {
113             if (TxtNam.Text != "")
114             {
115                 SqlCommand cmd = new SqlCommand();
116                 cmd.Connection = con;
117                 cmd.CommandText = "SELECT * FROM Book where Title Like '" + TxtTitle.Text + "%'";
118                 SqlDataAdapter da = new SqlDataAdapter(cmd);
119                 DataSet ds = new DataSet();
120                 da.Fill(ds);
121                 GridBook.DataSource = ds.Tables[0];
122
123             }
124         }
125
        1 reference
126         private void BtnRef_Click(object sender, EventArgs e)
127         {
128             TxtNam.Clear();
129             PanelEdit.Visible = false;
130         }
131
        1 reference
132         private void Btnupdt_Click(object sender, EventArgs e)
133         {
134             try
135             {
136                 if (MessageBox.Show("Data is going to be updated.Confirm?", "sucess", MessageBoxButtons.OKCancel, MessageBoxIcon.Question) == DialogResult.OK)
137                 {
138                     Int64 isbn = Int64.Parse(TxtIsbn.Text);
139                     noofcopies = Int64.Parse(Txtcons.Text);
140                     con.Open();
141
142                     SqlCommand cmd = new SqlCommand();
143                     cmd.Connection = con;
144                     cmd.CommandText = "UPDATE Book SET Author = @Author, Title = @Title, Publisher = @Publisher, Genre = @Genre, Permission = @Permission, Shelf = @Shelf, NumberOfCopies = @NumberOfCopies WHERE ISBN = @ISBN";
145                     cmd.Parameters.AddWithValue("@ISBN", isbn);
146                     cmd.Parameters.AddWithValue("@Author", TxtAuthor.Text);
147                     cmd.Parameters.AddWithValue("@Title", TxtTitle.Text);
148                     cmd.Parameters.AddWithValue("@Publisher", TxtPublisher.Text);
149                     cmd.Parameters.AddWithValue("@Genre", TxtGenre.Text);
150                     cmd.Parameters.AddWithValue("@Permission", TxtPermission.Text);
151                     cmd.Parameters.AddWithValue("@Shelf", TxtShelf.Text);
152                     cmd.Parameters.AddWithValue("@NumberOfCopies", noofcopies);
153
154                     cmd.ExecuteNonQuery();
155                     con.Close();
156                     SqlCommand refreshCmd = new SqlCommand("SELECT * FROM Book", con);
157                     SqlDataAdapter da = new SqlDataAdapter(refreshCmd);
158                     DataSet ds = new DataSet();
159                     da.Fill(ds);
160                     if (ds.Tables.Count > 0)
161                     {
162                         GridBook.DataSource = ds.Tables[0];
163                     }
164                 }
165             }
```

```
166             catch (Exception ex)
167             {
168                 MessageBox.Show(ex.Message);
169             }
170         }
171
        1 reference
172         private void Btndel_Click(object sender, EventArgs e)
173         {
174             if (MessageBox.Show("Data is going to be updated.Confirm?", "sucess", MessageBoxButtons.OKCancel, MessageBoxIcon.Warning) == DialogResult.OK)
175             {
176                 SqlCommand cmd = new SqlCommand();
177                 cmd.Connection = con;
178                 con.Open();
179                 cmd.CommandText = "delete from Book  where Isbn = '" + ISBN + "'";
180                 SqlDataAdapter da = new SqlDataAdapter(cmd);
181                 DataSet ds = new DataSet();
182                 da.Fill(ds);
183                 con.Close();
184             }
185         }
186     }
187 }
188
```

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.Common;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace RAD_Library_Management_System
{
    public partial class FormBooK : Form
    {
        public FormBooK()
        {
            InitializeComponent();
        }

        SqlConnection con = new SqlConnection("Data Source=DESKTOP-
3J22NMI\\SQLEXPRESS;Initial Catalog=LibraryA;Integrated Security=True");
```

```
private void BackN_Click(object sender, EventArgs e)
{
    this.Close();
}

private void MinimizeN_Click(object sender, EventArgs e)
{
    this.WindowState = FormWindowState.Minimized;
}

private void CLoseN_Click(object sender, EventArgs e)
{
    Application.Exit();
}

private void BtnSearch_Click(object sender, EventArgs e)
{
    string searchTitle = TxtNam.Text.Trim();
    SqlCommand cmd = new SqlCommand();
    cmd.Connection = con;
    cmd.CommandText = "SELECT * FROM Book where Title = '" + searchTitle + "'";
    SqlDataAdapter da = new SqlDataAdapter(cmd);
    DataSet ds = new DataSet();
    da.Fill(ds);
    GridBook.DataSource = ds.Tables[0];
    PanelEdit.Visible = true;

}

private void FormBooK_Load(object sender, EventArgs e)
{
    SqlCommand cmd = new SqlCommand();
    cmd.Connection = con;
    cmd.CommandText = "SELECT * FROM Book ";
    SqlDataAdapter da = new SqlDataAdapter(cmd);
    DataSet ds = new DataSet();
    da.Fill(ds);
    GridBook.DataSource = ds.Tables[0];
    PanelEdit.Visible = false;
}

Int64 ISBN;
Int64 rowid;
Int64 noofcopies;
private void GridBook_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
```

```
        try
        {
            if (GridBook.Rows[e.RowIndex].Cells[e.ColumnIndex].Value != null)
            {
                ISBN = Int64.Parse(GridBook.Rows[e.RowIndex].Cells[0].Value.ToString());
                noofcopies =
Int64.Parse(GridBook.Rows[e.RowIndex].Cells[0].Value.ToString());
                PanelEdit.Visible = true;
                SqlCommand cmd = new SqlCommand();
                cmd.Connection = con;
                cmd.CommandText = "SELECT * FROM Book WHERE ISBN = @ISBN";
                cmd.Parameters.AddWithValue("@ISBN", ISBN);
                SqlDataAdapter da = new SqlDataAdapter(cmd);
                DataSet ds = new DataSet();
                da.Fill(ds);
                if (ds.Tables.Count > 0 && ds.Tables[0].Rows.Count > 0)
                {
                    TxtIsbn.Text = ds.Tables[0].Rows[0]["ISBN"].ToString();
                    TxtTitle.Text = ds.Tables[0].Rows[0]["Title"].ToString();
                    TxtGenre.Text = ds.Tables[0].Rows[0]["Genre"].ToString();
                    TxtNocopiew.Text = ds.Tables[0].Rows[0]["NumberOfCopies"].ToString();
                    TxtPerMission.Text = ds.Tables[0].Rows[0]["Permission"].ToString();
                    TxtShwlf.Text = ds.Tables[0].Rows[0]["Shelf"].ToString();
                    TxtPublisher.Text = ds.Tables[0].Rows[0]["Publisher"].ToString();
                    TxtAuthor.Text = ds.Tables[0].Rows[0]["Author"].ToString();
                }
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show("An error occurred: " + ex.Message);
        }
    }
    private void TxtNam_TextChanged(object sender, EventArgs e)
    {
        if (TxtNam.Text != "")
        {
            SqlCommand cmd = new SqlCommand();
            cmd.Connection = con;
            cmd.CommandText = "SELECT * FROM Book where Title Like '" + TxtTitle.Text
+ "%'";
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataSet ds = new DataSet();
            da.Fill(ds);
            GridBook.DataSource = ds.Tables[0];
        }
```

```
        }

        private void BtnRef_Click(object sender, EventArgs e)
        {
            TxtNam.Clear();
            PanelEdit.Visible = false;
        }
        private void Btnupdt_Click(object sender, EventArgs e)
        {
            try
            {
                if (MessageBox.Show("Data is going to be updated.Confirm?", "sucess",
MessageBoxButtons.OKCancel, MessageBoxIcon.Question) == DialogResult.OK)
                {
                    Int64 isbn = Int64.Parse(TxtIsbn.Text);
                    noofcopies = Int64.Parse(Txtcono.Text);
                    con.Open();
                    SqlCommand cmd = new SqlCommand();
                    cmd.Connection = con;
                    cmd.CommandText = "UPDATE Book SET Author = @Author, Title = @Title,
Publisher = @Publisher, Genre = @Genre, Permission = @Permission, Shelf = @Shelf,
NumberOfCopies = @NumberOfCopies WHERE ISBN = @ISBN";
                    cmd.Parameters.AddWithValue("@ISBN", isbn);
                    cmd.Parameters.AddWithValue("@Author", TxtAuthor.Text);
                    cmd.Parameters.AddWithValue("@Title", TxtTitle.Text);
                    cmd.Parameters.AddWithValue("@Publisher", TxtPublisher.Text);
                    cmd.Parameters.AddWithValue("@Genre", TxtGenre.Text);
                    cmd.Parameters.AddWithValue("@Permission", TxtPerMission.Text);
                    cmd.Parameters.AddWithValue("@Shelf", TxtShwlf.Text);
                    cmd.Parameters.AddWithValue("@NumberOfCopies", noofcopies);
                    cmd.ExecuteNonQuery();
                    con.Close();
                    SqlCommand refreshCmd = new SqlCommand("SELECT * FROM Book", con);
                    SqlDataAdapter da = new SqlDataAdapter(refreshCmd);
                    DataSet ds = new DataSet();
                    da.Fill(ds);
                    if (ds.Tables.Count > 0)
                    {
                        GridBook.DataSource = ds.Tables[0];
                    }
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show(ex.Message);
            }
```

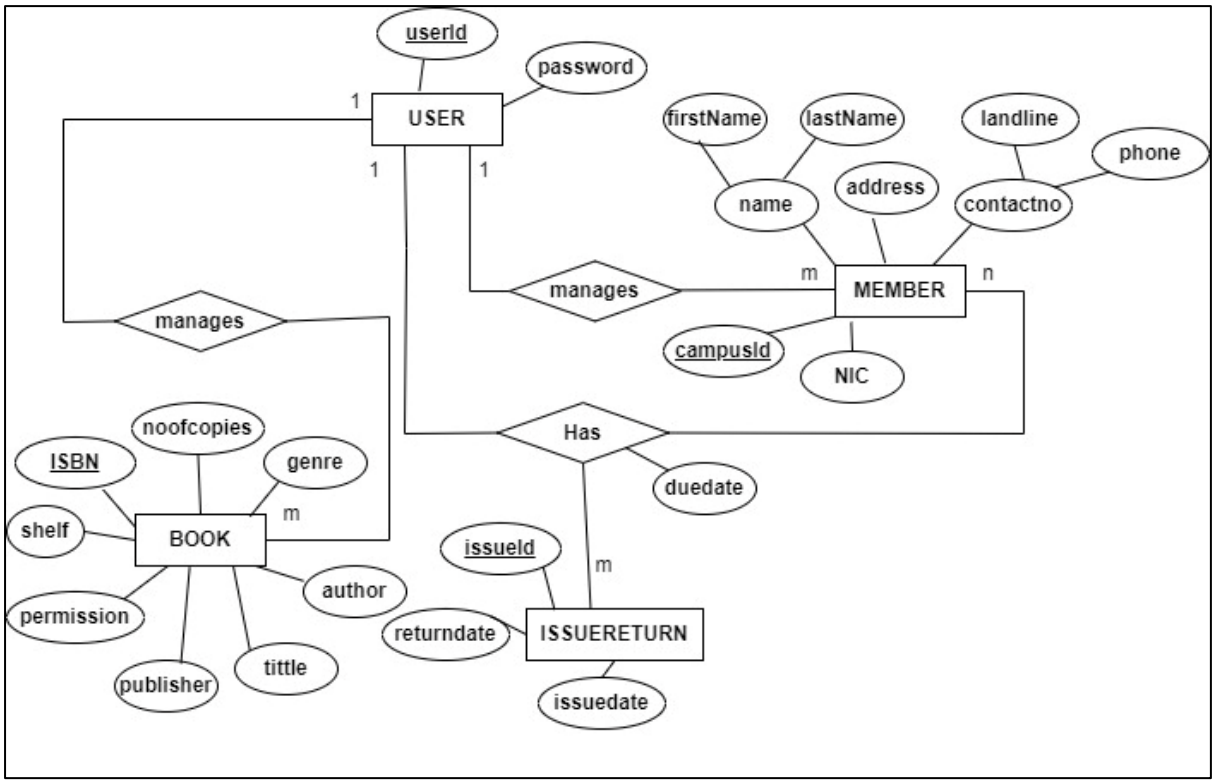
```
        }

        private void Btndel_Click(object sender, EventArgs e)
        {
            if (MessageBox.Show("Data is going to be updated.Confirm?", "sucess",
MessageBoxButtons.OKCancel, MessageBoxIcon.Warning) == DialogResult.OK)
            {
                SqlCommand cmd = new SqlCommand();
                cmd.Connection = con;
                con.Open();
                cmd.CommandText = "delete from Book  where Isbn = '" + ISBN + "'";
                SqlDataAdapter da = new SqlDataAdapter(cmd);
                DataSet ds = new DataSet();
                da.Fill(ds);
                con.Close();
            }
        }
    }
}
```

3. Database

LibraryA.sql - DESK...ndya Senarath (51))*

```sql
create database LibraryA

use LibraryA

--userLogin table
CREATE TABLE [dbo].[UserLogin] (
    [userId] NVARCHAR (50) NULL,
    [pwsd]   NVARCHAR (12) NULL
);

--Book table
CREATE TABLE Book(
    ISBN BIGINT IDENTITY(483948274987,1) PRIMARY KEY NOT NULL,
    Author nvarchar(100) NOT NULL,
    Title nvarchar(100) NOT NULL,
    Publisher nvarchar(100)NOT NULL,
    Genre nvarchar(50) NOT NULL,
    Permission nvarchar(50) NOT NULL,
    shelf nvarchar(50) NOT NULL,
    NumberOfCopies int NOT NULL
);


--Memeber table
CREATE TABLE LMember
(
    CampusId nvarchar (6) primary key NOT NULL,
    NIC nvarchar(255),
    FirstName nvarchar(255),
    Lastname nvarchar(255),
    land int ,
    phone int NOT NULL ,
    mAddress Nvarchar(255),
);

--return an dissue books
CREATE TABLE  ReturnIssue
(
    IssueId INT IDENTITY(1,1) PRIMARY KEY,
    CampusId NVARCHAR(6) NOT NULL,
    ISBN BIGINT NOT NULL,
    IssueDate DATE NOT NULL,
    ReturnDate DATE,
    FOREIGN KEY (CampusId) REFERENCES LMember(CampusId),
    FOREIGN KEY (ISBN) REFERENCES Book(ISBN)
);
```

- **Database Script**

```
Create database LibraryA

use LibraryA

--user Login table
CREATE TABLE [dbo].[UserLogin] (
   [userId] NVARCHAR (50) NULL,
   [pwsd]   NVARCHAR (12) NULL
);

--Book table
CREATE TABLE Book(
   ISBN BIGINT IDENTITY(483948274987,1) PRIMARY KEY NOT NULL,
   Author nvarchar(100) NOT NULL,
   Title nvarchar(100) NOT NULL,
   Publisher nvarchar(100)NOT NULL,
   Genre nvarchar(50) NOT NULL,
   Permission nvarchar(50) NOT NULL,
         shelf nvarchar(50) NOT NULL,
   NumberOfCopies int NOT NULL
);

--Memeber table
CREATE TABLE LMember
(
         CampusId nvarchar (6) primary key NOT NULL,
         NIC nvarchar(255),
         FirstName nvarchar(255),
         Lastname nvarchar(255),
         land int ,
         phone int NOT NULL ,
         mAddress Nvarchar(255),
);

--return an dissue books
CREATE TABLE ReturnIssue
(
   IssueId INT IDENTITY (1,1) PRIMARY KEY,
   CampusId NVARCHAR (6) NOT NULL,
   bTitle nvarchar(100) NOT NULL,
   IssueDate DATE NOT NULL,
   ReturnDate DATE,
   FOREIGN KEY (CampusId) REFERENCES LMember(CampusId),

);
```

4. Revised ER Diagram